

Physical Chemistry (II)

Lecture 07

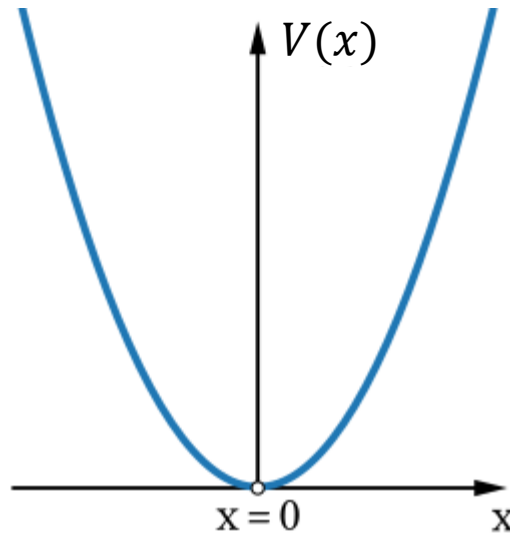
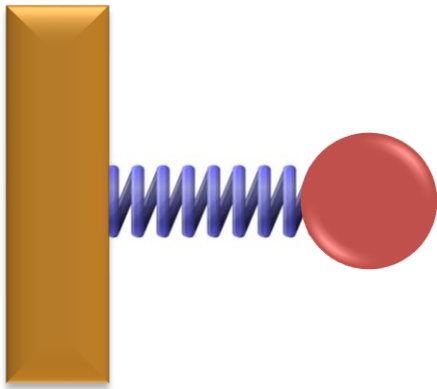
Simple Harmonic Oscillator

Rigid Rotor



Last class

Simple Harmonic Oscillator



$$V(x) = \frac{1}{2} kx^2$$

$$F(x) = -kx$$

Quantum Mechanics

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \mathcal{V}(\hat{x}) \right] \psi(x) = E\psi(x)$$
$$\left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} kx^2 \right) \psi(x) = E\psi(x)$$

No simple method to solve this equation

But we “know” the solutions

Solutions

- Quantum number: $v = 0, 1, 2, \dots$
- Energy

$$E_v = \frac{h}{2\pi} \left(\frac{k}{m} \right)^{1/2} \left(v + \frac{1}{2} \right) = h\nu \left(v + \frac{1}{2} \right)$$

- Wavefunction

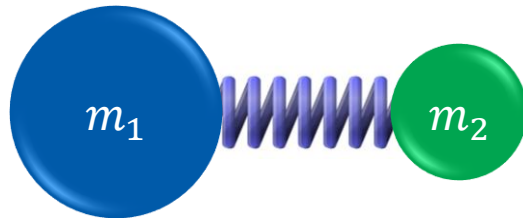
$$\psi_v(x) = \frac{(\alpha/\pi)^{1/4}}{(2^v v!)^{1/2}} \underbrace{H_v(\alpha^{1/2} x)}_{\text{Hermite polynomial}} e^{-\alpha x^2/2}$$

where $\alpha = \sqrt{mk}/\hbar$

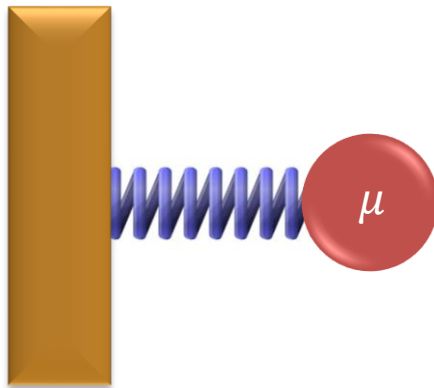


Last class

Two Connected Bodies



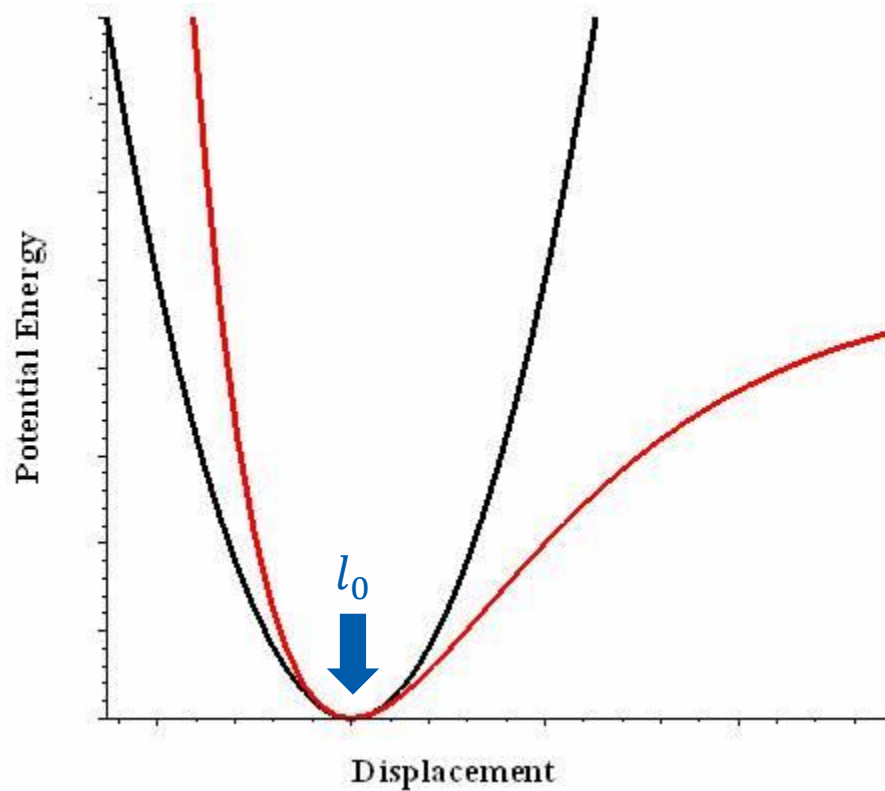
Two-body problem



One-body problem
(SHO)

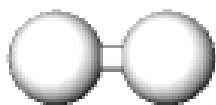
Last class

Approximation to Any Potential

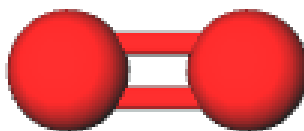


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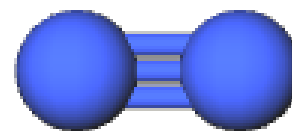
Model for Molecular Vibration



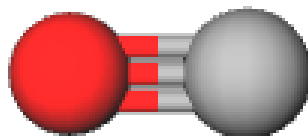
H_2



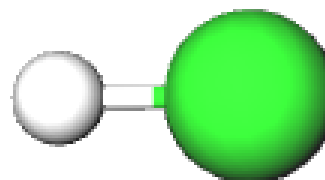
O_2



N_2



CO



HCl

Selection Rules

- **Selection rules:** some transitions are allowed, and the others are prohibited by QM
- Selection rule 1 for SHO

$$\Delta v = \pm 1$$

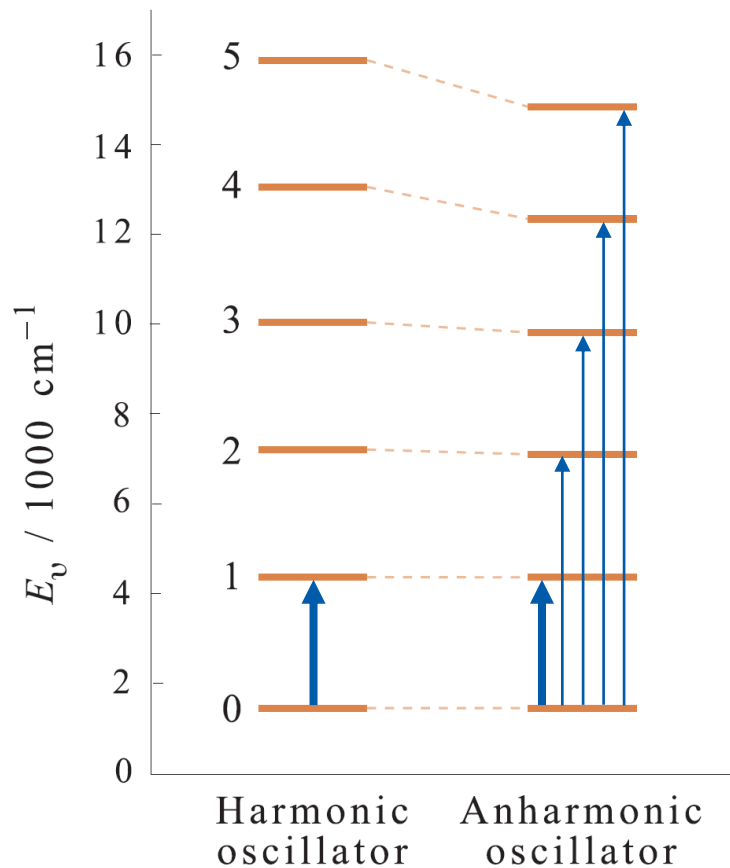
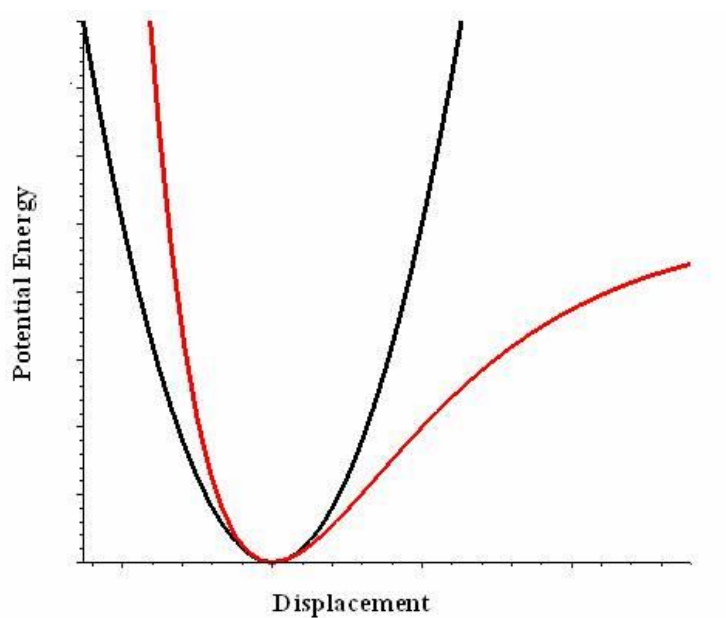
- Selection rule 2 for SHO

$$\frac{d\mu}{dx} \neq 0$$

“Dipole moment of the molecule must vary during vibration”

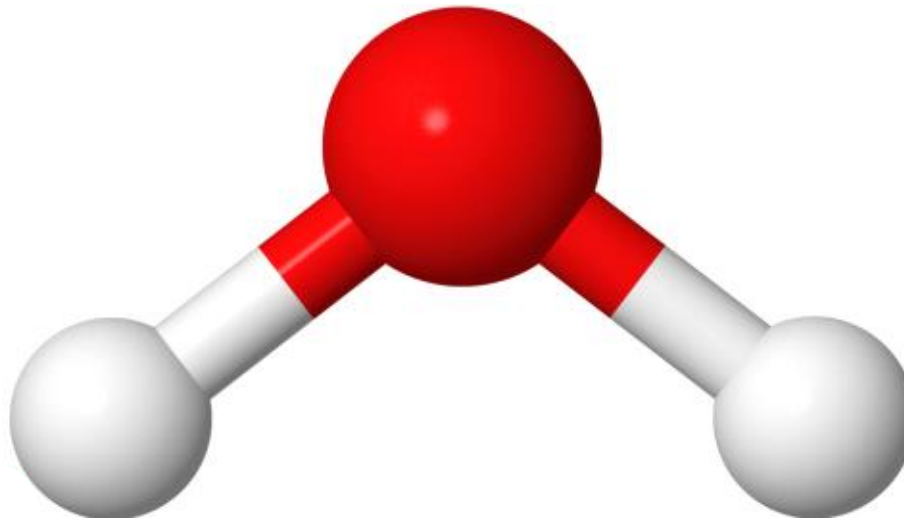
Last class

Anharmonicity



RS 5.3
M 5.11

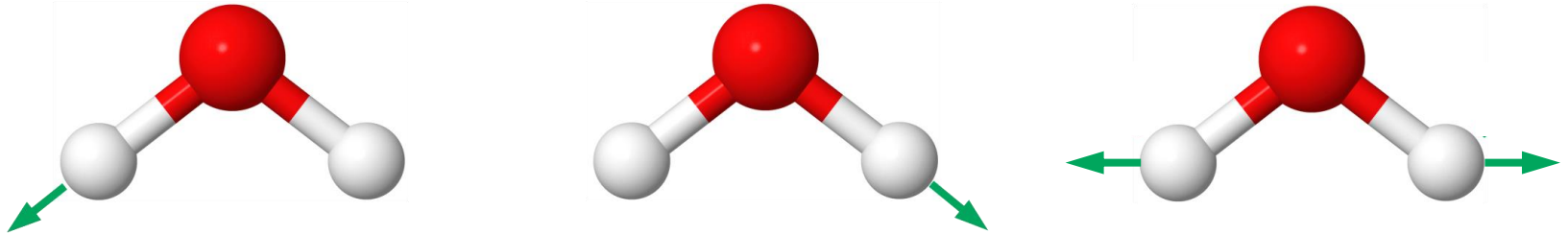
Polyatomic Molecules



Vibrations?

RS 5.3
M 5.11

Vibration of Each Bond?



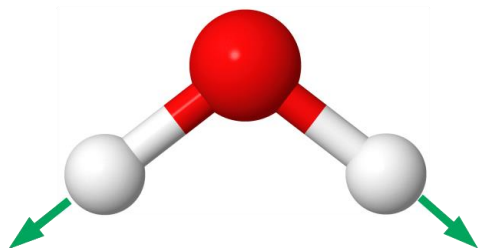
Center of mass is not fixed!
(coupled with translation)

Molecular Motions

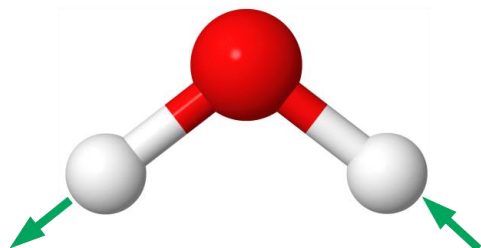
- Degree of freedom = $3N$ (N : number of atoms)
- Translation: 3
- Rotation: 2 (linear) or 3 (non-linear)
- Vibration: $3N - 5$ (linear) or $3N - 6$ (non-linear)
→ Number of possible vibrational modes

RS 5.3
M 5.11

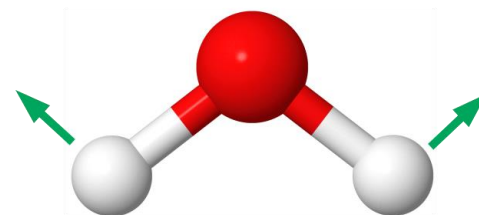
Normal Modes



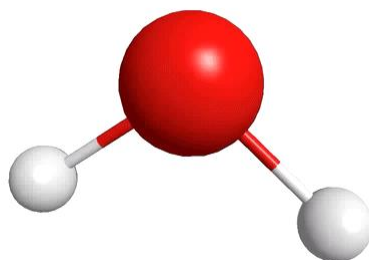
symmetric stretching



asymmetric stretching



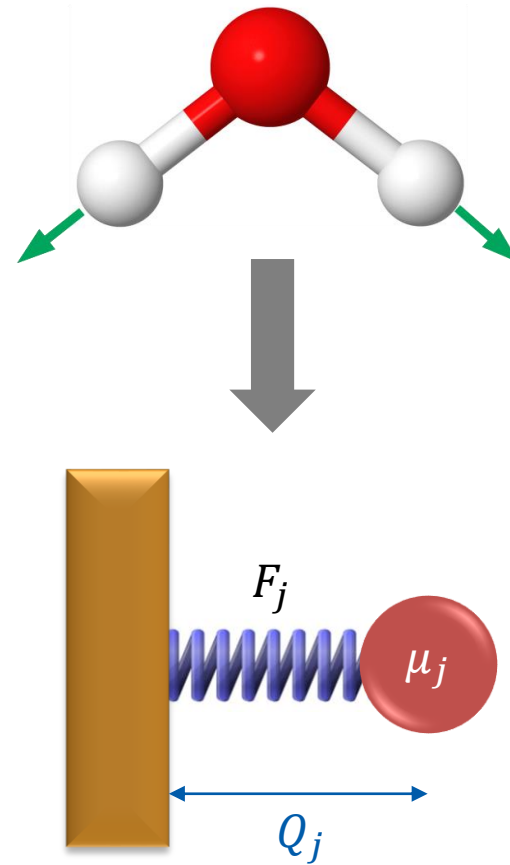
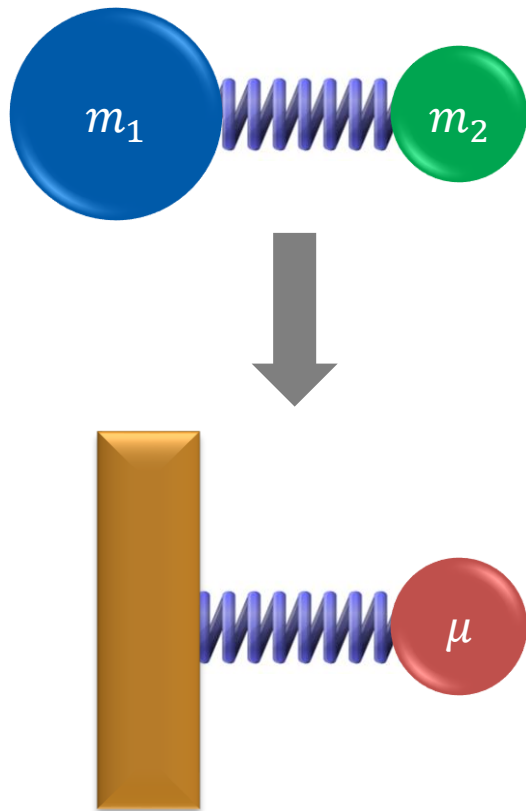
bending



(Images: https://strawberryfields.ai/phononics/_images/water.png
<https://thepaperccny.com/2022/02/28/harmony-of-the-ocean/>)

RS 5.3
M 5.11

Conversion



Quantum Mechanics

$$V(Q_1, Q_2, \dots, Q_{N_{\text{vib}}}) = \frac{1}{2} \sum_{j=1}^{N_{\text{vib}}} F_j Q_j^2$$

$$\hat{H}_{\text{vib}} = \sum_{j=1}^{N_{\text{vib}}} \underbrace{\left(-\frac{\hbar^2}{2\mu_j} \frac{\partial^2}{\partial Q_j^2} + \frac{1}{2} F_j Q_j^2 \right)}_{\hat{H}_{\text{vib},j}}$$

$$\begin{aligned} \hat{H}_{\text{vib}} \psi(Q_1, Q_2, \dots, Q_{N_{\text{vib}}}) &= E \psi(Q_1, Q_2, \dots, Q_{N_{\text{vib}}}) \\ &= (\hat{H}_{\text{vib},1} + \hat{H}_{\text{vib},2} + \dots + \hat{H}_{\text{vib},N_{\text{vib}}}) \psi(Q_1, Q_2, \dots, Q_{N_{\text{vib}}}) \end{aligned}$$

Separation of Variables

$$(\hat{H}_{\text{vib},1} + \hat{H}_{\text{vib},2} + \cdots + \hat{H}_{\text{vib},N_{\text{vib}}})\psi(Q_1, Q_2, \cdots, Q_{N_{\text{vib}}}) = E\psi$$

Use separation of variables:

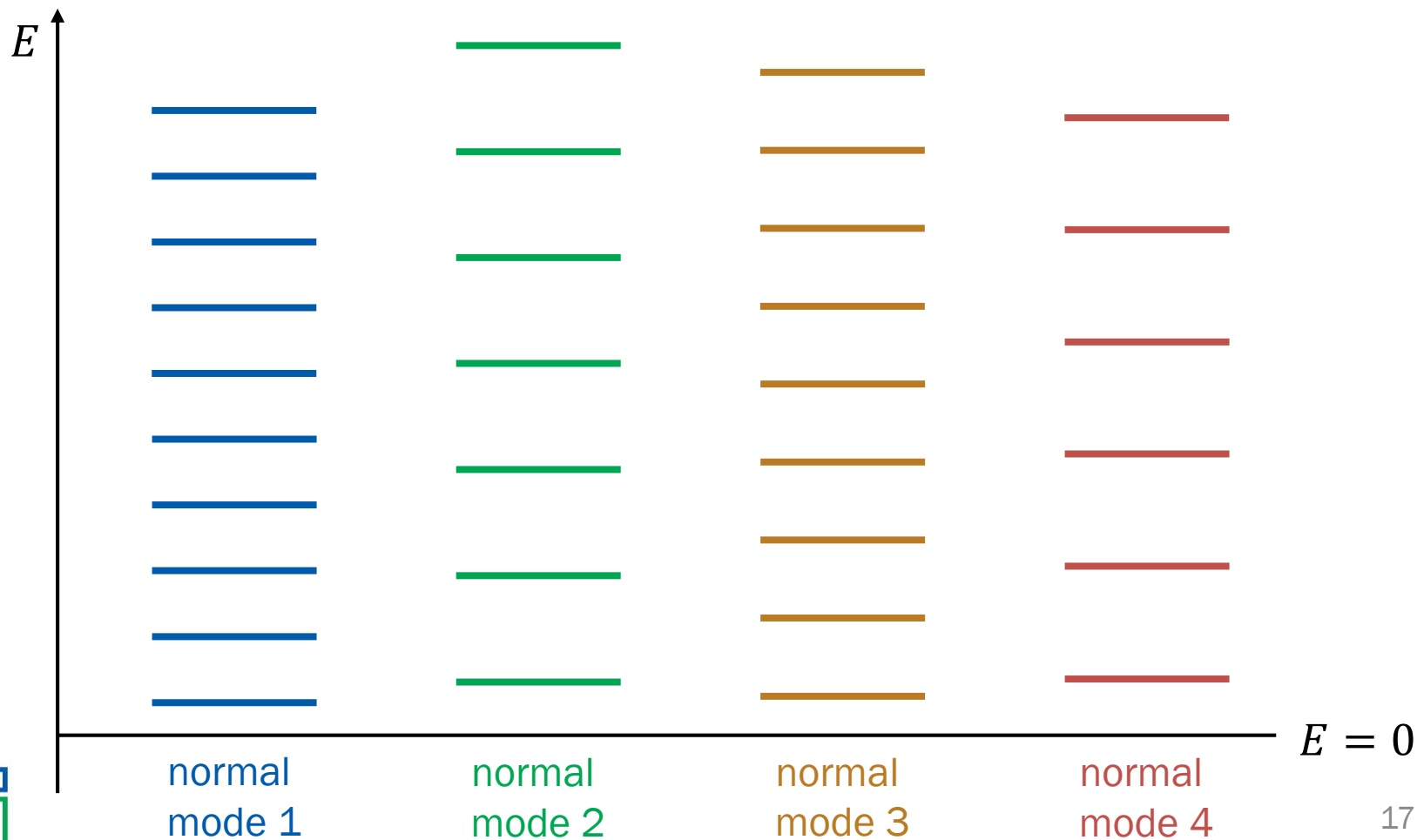
$$\psi(Q_1, Q_2, \cdots, Q_{N_{\text{vib}}}) = \psi_{\text{vib},1}(Q_1)\psi_{\text{vib},2}(Q_2) \cdots \psi_{\text{vib},N}(Q_{N_{\text{vib}}})$$

$$E_{\text{vib}} = \sum_{j=1}^{N_{\text{vib}}} E_{\text{vib},j} = \sum_{j=1}^{N_{\text{vib}}} h\nu_j \left(v_j + \frac{1}{2} \right)$$

$$(\text{where } \nu_j = \sqrt{F_j/\mu_j})$$

RS 5.3
M 5.11

Vibration of Polyatomics



Selection Rules

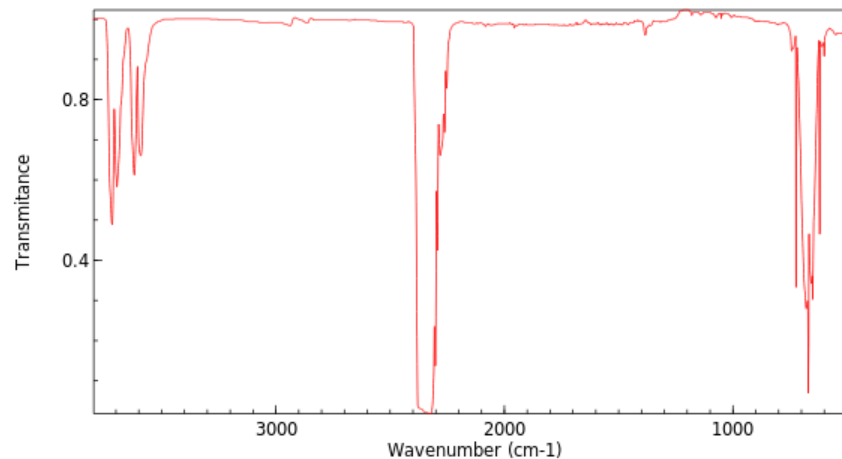
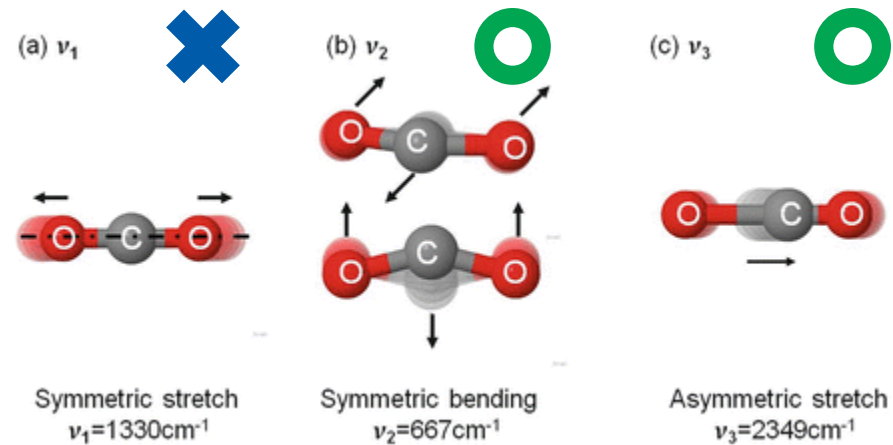
$$\Delta v_j = \pm 1$$

$$\frac{d\mu}{dQ_j} \neq 0$$

Dipole moment of the molecule must vary during a vibration

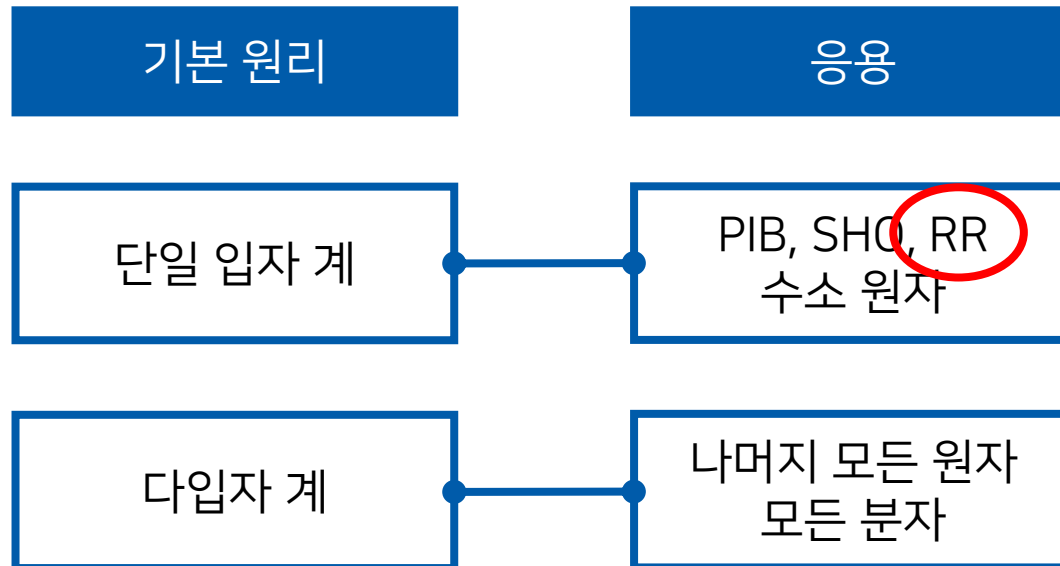
- **Infrared active:** the normal mode whose $d\mu/dQ_j \neq 0$
- **Infrared inactive:** the normal mode whose $d\mu/dQ_j = 0$

Example: IR Spectrum of CO₂



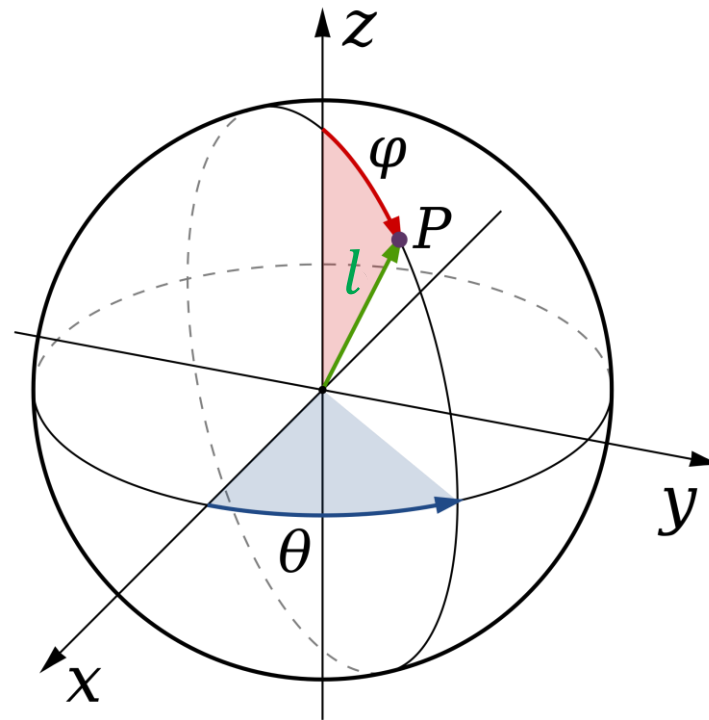
(Image: Hansen et al., "Polyatomic Molecular Spectra," *Spectroscopy and Optical Diagnostics for Gases*, 2016) 19

Where Are We?



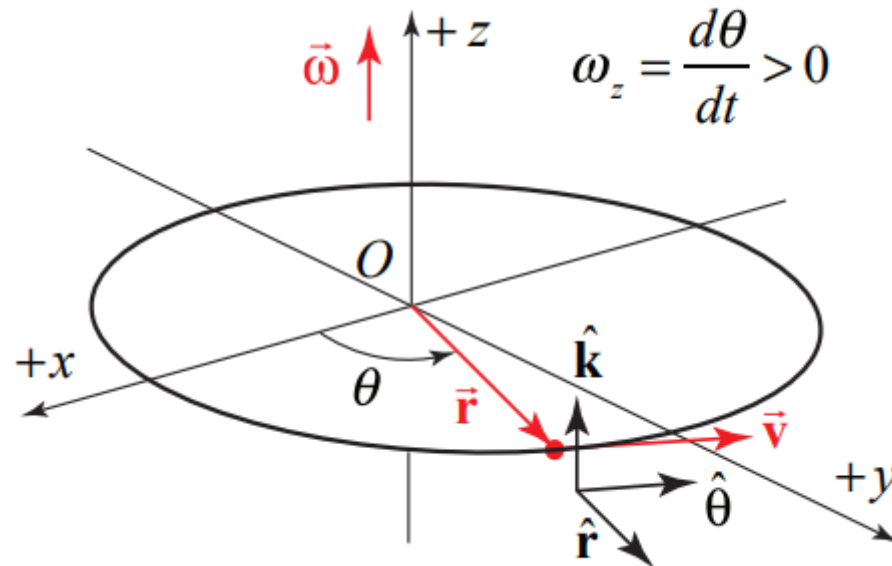
RS 4.3
M 6.1

Rigid Rotor Model (a.k.a. Particle on a Sphere)



(Image: [https://commons.wikimedia.org/wiki/File:Spherical_Coordinates_\(Colatitude,_Longitude\).svg](https://commons.wikimedia.org/wiki/File:Spherical_Coordinates_(Colatitude,_Longitude).svg))

Angular Velocity



$$\vec{\omega} = \frac{\vec{r} \times \vec{v}}{|\vec{r}|^2}$$

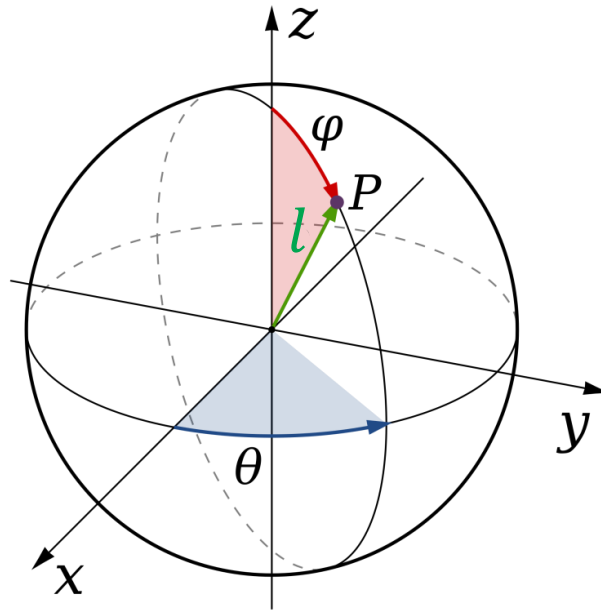
Angular Momentum

- Momentum $\vec{p} = m\vec{v}$: conserved in linear motion
- Angular momentum $\vec{L} = I\vec{\omega}$: conserved in circular motion
where the moment of inertia $I = m|r|^2$

Linear vs. Rotational Motion

Linear motion	Rotational motion
Mass m	Moment of inertia I
Displacement $\vec{x}$	Angular displacement $\vec{\theta}$
Velocity $\vec{v}$	Angular velocity $\vec{\omega}$
Acceleration $\vec{a}$	Angular acceleration $\vec{\alpha}$
Momentum $\vec{p} = m\vec{v}$	Angular momentum $\vec{L} = I\vec{\omega}$
Force $\vec{F} = m\vec{a} = \frac{d\vec{p}}{dt}$	Force $\vec{F} = I\vec{\alpha} = \frac{d\vec{L}}{dt}$
Kinetic energy $T = \frac{1}{2}mv^2$	Kinetic energy $T = \frac{1}{2}I\omega^2$
If $\vec{F}_{\text{ext}} = 0, \frac{d\vec{p}}{dt} = 0$	If $\vec{F}_{\text{ext}} = 0, \frac{d\vec{L}}{dt} = 0$

Hamiltonian for Rotation



$$\hat{H} = \hat{T} + \hat{V} = \frac{\hat{L}^2}{2I}$$

Angular Momenta in CM

$$\text{As } \vec{\omega} = (\vec{r} \times \vec{v})/|r|^2,$$

$$\begin{aligned}\vec{L} &= m|r|^2 \cdot \frac{\vec{r} \times \vec{v}}{|r|^2} = \vec{r} \times (m\vec{v}) = \vec{r} \times \vec{p} \\ &= (yp_z - zp_y, zp_x - xp_z, xp_y - yp_x)\end{aligned}$$

$$L_x = yp_z - zp_y$$

$$L_y = zp_x - xp_z$$

$$L_z = xp_y - yp_x$$

Angular Momenta in QM

$$\hat{L}_x = y\hat{p}_z - z\hat{p}_y = -i\hbar \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right)$$

$$\hat{L}_y = z\hat{p}_x - x\hat{p}_z = -i\hbar \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right)$$

$$\hat{L}_z = x\hat{p}_y - y\hat{p}_x = -i\hbar \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right)$$

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$$

RS 4.6

M 6.8

Commutators

$$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z$$

$$[\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_x$$

$$[\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y$$

$$[\hat{L}^2, \hat{L}_x] = 0$$

$$[\hat{L}^2, \hat{L}_y] = 0$$

$$[\hat{L}^2, \hat{L}_z] = 0$$

Not in text

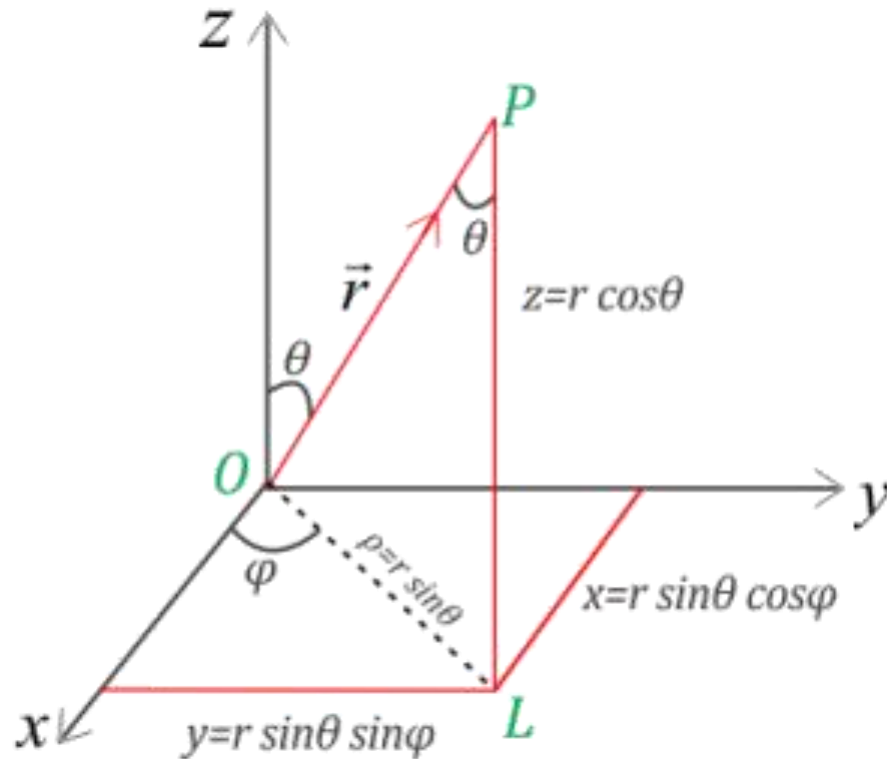
Hamiltonian?

$$\hat{H} = \frac{\hat{L}^2}{2I} = \frac{\hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2}{2I}$$

$$\hat{H} = -\frac{\hbar^2}{2I} \left[(y^2 + z^2) \frac{\partial^2}{\partial x^2} + (z^2 + x^2) \frac{\partial^2}{\partial y^2} + (x^2 + y^2) \frac{\partial^2}{\partial z^2} \right. \\ \left. - 2 \left(x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} + z \frac{\partial}{\partial z} + yz \frac{\partial^2}{\partial y \partial z} + zx \frac{\partial^2}{\partial z \partial x} + xy \frac{\partial^2}{\partial x \partial y} \right) \right]$$

Can we solve $\hat{H}\psi = E\psi$?

Spherical Coordinates



$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \cos^{-1}(z/r)$$

$$\phi = \tan^{-1}(y/x)$$

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$\hat{L}_i$ in Spherical Coordinates

$$\hat{L}_x = -i\hbar \left(-\sin \phi \frac{\partial}{\partial \theta} - \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right)$$

$$\hat{L}_y = -i\hbar \left(\cos \phi \frac{\partial}{\partial \theta} - \cot \theta \sin \phi \frac{\partial}{\partial \phi} \right)$$

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}$$

RS 4.6

M 6.8

$\hat{L}_x^2$ Operator?

$$\hat{L}_x = i\hbar \left(\sin \phi \frac{\partial}{\partial \theta} + \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right)$$

$$\begin{aligned}\hat{L}_x^2 f &= (i\hbar)^2 \left(\sin \phi \frac{\partial}{\partial \theta} + \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right) \left(\sin \phi \frac{\partial}{\partial \theta} + \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right) f \\&= -\hbar^2 \left(\sin \phi \frac{\partial}{\partial \theta} + \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right) \left(\sin \phi \frac{\partial f}{\partial \theta} + \cot \theta \cos \phi \frac{\partial f}{\partial \phi} \right) \\&= -\hbar^2 \left[\sin \phi \frac{\partial}{\partial \theta} \left(\sin \phi \frac{\partial f}{\partial \theta} + \cot \theta \cos \phi \frac{\partial f}{\partial \phi} \right) \right. \\&\quad \left. + \cot \theta \cos \phi \frac{\partial}{\partial \phi} \left(\sin \phi \frac{\partial f}{\partial \theta} + \cot \theta \cos \phi \frac{\partial f}{\partial \phi} \right) \right]\end{aligned}$$

RS 4.6

M 6.8

Almost There ...

$$\begin{aligned}
 \hat{L}_x^2 f &= -\hbar^2 \left[\sin \phi \frac{\partial}{\partial \theta} \left(\sin \phi \frac{\partial f}{\partial \theta} + \cot \theta \cos \phi \frac{\partial f}{\partial \phi} \right) \right. \\
 &\quad \left. + \cot \theta \cos \phi \frac{\partial}{\partial \phi} \left(\sin \phi \frac{\partial f}{\partial \theta} + \cot \theta \cos \phi \frac{\partial f}{\partial \phi} \right) \right] \\
 &= -\hbar^2 \left(\sin^2 \phi \frac{\partial^2 f}{\partial \theta^2} - \csc^2 \theta \sin \phi \cos \phi \frac{\partial f}{\partial \phi} + \cot \theta \sin \phi \cos \phi \frac{\partial^2 f}{\partial \theta \partial \phi} \right. \\
 &\quad \left. + \cot \theta \cos^2 \phi \frac{\partial f}{\partial \theta} + \cot \theta \sin \phi \cos \phi \frac{\partial^2 f}{\partial \phi \partial \theta} \right. \\
 &\quad \left. - \cot^2 \theta \sin \phi \cos \phi \frac{\partial f}{\partial \phi} + \cot^2 \theta \cos^2 \phi \frac{\partial^2 f}{\partial \phi^2} \right)
 \end{aligned}$$

RS 4.6

M 6.8

Final Form

$$\hat{L}_x^2 f = -\hbar^2 \left(\sin^2 \phi \frac{\partial^2 f}{\partial \theta^2} - \csc^2 \theta \sin \phi \cos \phi \frac{\partial f}{\partial \phi} + \cot \theta \sin \phi \cos \phi \frac{\partial^2 f}{\partial \theta \partial \phi} \right. \\ \left. + \cot \theta \cos^2 \phi \frac{\partial f}{\partial \theta} + \cot \theta \sin \phi \cos \phi \frac{\partial^2 f}{\partial \phi \partial \theta} \right. \\ \left. - \cot^2 \theta \sin \phi \cos \phi \frac{\partial f}{\partial \phi} + \cot^2 \theta \cos^2 \phi \frac{\partial^2 f}{\partial \phi^2} \right)$$

$$\hat{L}_x^2 f = -\hbar^2 \left[\sin^2 \phi \frac{\partial^2 f}{\partial \theta^2} + \cot \theta \cos^2 \phi \frac{\partial f}{\partial \theta} + 2 \cot \theta \sin \phi \cos \phi \frac{\partial^2 f}{\partial \theta \partial \phi} \right. \\ \left. + \cot^2 \theta \cos^2 \phi \frac{\partial^2 f}{\partial \phi^2} - (\csc^2 \theta + \cot^2 \theta) \sin \phi \cos \phi \frac{\partial f}{\partial \phi} \right]$$

RS 4.6

M 6.8

$$\hat{L}_x^2 + \hat{L}_y^2$$

$$\hat{L}_x^2 f = -\hbar^2 \left[\sin^2 \phi \frac{\partial^2 f}{\partial \theta^2} + \cot \theta \cos^2 \phi \frac{\partial f}{\partial \theta} + 2 \cot \theta \sin \phi \cos \phi \frac{\partial^2 f}{\partial \theta \partial \phi} \right. \\ \left. + \cot^2 \theta \cos^2 \phi \frac{\partial^2 f}{\partial \phi^2} - (\csc^2 \theta + \cot^2 \theta) \sin \phi \cos \phi \frac{\partial f}{\partial \phi} \right]$$

$$\hat{L}_y^2 f = -\hbar^2 \left[\cos^2 \phi \frac{\partial^2 f}{\partial \theta^2} + \cot \theta \sin^2 \phi \frac{\partial f}{\partial \theta} - 2 \cot \theta \sin \phi \cos \phi \frac{\partial^2 f}{\partial \theta \partial \phi} \right. \\ \left. + \cot^2 \theta \sin^2 \phi \frac{\partial^2 f}{\partial \phi^2} + (\csc^2 \theta + \cot^2 \theta) \sin \phi \cos \phi \frac{\partial f}{\partial \phi} \right]$$

$$(\hat{L}_x^2 + \hat{L}_y^2)f = -\hbar^2 \left(\frac{\partial^2 f}{\partial \theta^2} + \cot \theta \frac{\partial f}{\partial \theta} + \cot^2 \theta \frac{\partial^2 f}{\partial \phi^2} \right)$$

RS 4.6

M 6.8

$$\hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$$

$$(\hat{L}_x^2 + \hat{L}_y^2)f = -\hbar^2 \left(\frac{\partial^2 f}{\partial \theta^2} + \cot \theta \frac{\partial f}{\partial \theta} + \cot^2 \theta \frac{\partial^2 f}{\partial \phi^2} \right)$$

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}$$

$$\hat{L}_z^2 = -\hbar^2 \frac{\partial^2}{\partial \phi^2}$$

Hence,

$$(\hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2)f = -\hbar^2 \left[\frac{\partial^2 f}{\partial \theta^2} + \cot \theta \frac{\partial f}{\partial \theta} + (\cot^2 \theta + 1) \frac{\partial^2 f}{\partial \phi^2} \right]$$

Finally, $\hat{L}^2$

$$\begin{aligned}(\hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2)f &= -\hbar^2 \left[\frac{\partial^2 f}{\partial \theta^2} + \cot \theta \frac{\partial f}{\partial \theta} + (\cot^2 \theta + 1) \frac{\partial^2 f}{\partial \phi^2} \right] \\ &= -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2} \right]\end{aligned}$$

Hence,

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right]$$

Hamiltonian for Rigid Rotor

$$\hat{H} = \frac{\hat{L}^2}{2I} = \frac{\hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2}{2I}$$

$$\hat{H} = -\frac{\hbar^2}{2I} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right]$$

$$\hat{H}\psi(\theta, \phi) = E\psi(\theta, \phi)$$

$$-\frac{\hbar^2}{2I} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \psi(\theta, \phi) = E\psi(\theta, \phi)$$